Freight Rate Prediction Challenge — Machine Learning Technical Report

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TEMPORAL HOLDOUT RMSE
$40.73 / load
Sep–Oct 2025 ($0.0337/mi | R^2 = 0.9991)

TEMPORAL HOLDOUT MAPE
1.197%
Down from 6.563% Naive Baseline

UNSEEN CITIES SPATIAL R^2
0.9957 (LOO)
2D Gaussian RBF across 8 New Cities

CALENDAR & MARKET FIT
R^2 = 0.9970
304 Daily FEs + Quarter-End Surges

1. Data Exploration & Intentional Data Quality Issues Identified

Forensic inspection of the development dataset (`data/train_test.csv`, 48,000 loads from Jan 1 to Oct 31, 2025), the final prediction dataset (`data/validation.csv`, 12,000 loads from Nov 1 to Dec 31, 2025), and the fixed corridor benchmark (`data/december_chart_inputs.csv`, 31 loads from Dec 1 to Dec 31, 2025) revealed five critical data quality anomalies and distributional shifts engineered into the challenge:

Feature / Anomaly	Empirical Discovery & Forensic Evidence	Remediation & Modeling Rationale
1. `quote_signal` *Multi-Regime Leakage & Noise Trap*	Exhibits three incompatible regimes across time: • **Months 1, 2, 3, 6, 9 (Train):** Direct target leakage of rate-per-mile (`RPM = posted_rate / distance`), with `quote_signal ≈ RPM + N(0, 0.022²)` (96.3% within $0.05/mi). • **Months 4, 5, 7, 10 (Train):** Inverted target leakage where `quote_signal ≈ 4.15 − RPM + N(0, 0.030²)` (correlation flips to $r = -0.47$). • **Month 8 (Train) & Months 11–12 (Validation):** Pure Gaussian white noise `N(2.051, 0.220²)` with zero correlation to rate, distance, equipment, or weight, and completely absent from `december_chart_inputs.csv`.	**Dropped completely from all predictive models.** Any model trained on random splits or retaining `quote_signal` suffers severe accuracy collapse on `validation.csv` ($191.13 RMSE vs. $40.73 RMSE). Used solely offline during EDA to cross-verify that the 677 corrupted target outliers had normal underlying uncorrupted rates (mean = $2.21/mi).
2. `posted_rate` *Corrupted Target Outliers (1.41%)*	Evaluating `RPM = posted_rate / distance` reveals a sharp trimodal separation: • **Clean Domain:** 47,323 loads (98.59%) lie tightly in `[$1.576, $3.572]/mi`. • **Downward Corrupted:** 337 loads (0.70%) lie in `[$0.334, $1.253]/mi` (~0.30× normal), separated by an empty gap `($1.253, $1.576)/mi` with 0 rows. • **Upward Corrupted:** 340 loads (0.71%) lie in `[$4.058, $14.125]/mi` (~3.44× normal), separated by an empty gap `($3.572, $4.058)/mi` with 0 rows.	**Filtered training set to clean domain `1.35 ≤ RPM ≤ 3.80` (*N* = 47,323).** Removing the 677 corrupted target labels prevents 3–5× multiplicative outliers from biasing regression slopes, lane fixed effects, and log-normal variance smearing estimates.
3. `weight` *Sign-Flipped Negative & Missing Values*	• **Sign-Flipped Weights:** 292 train rows (0.61%) and 145 validation rows (1.21%) have negative values in `[−47,500, −5,000] lb` whose absolute values mirror the positive distribution `[5,000, 47,500] lb`. • **Missing Values:** 300 train rows (0.63%) and 165 validation rows (1.38%) are `NaN`.	**Applied `abs(weight)` and median imputation (31,493.5 lb) + `wt_na` flag.** Regression testing confirmed negative-weight loads follow the exact same rate curve as positive weights once sign-corrected ($\hat{\beta}_{neg_flag} = +0.0017$, statistically indistinguishable from zero).
4. `market_index` *Missing Values & EIV Attenuation*	Missing on 374 train rows (0.78%), 249 validation rows (2.08%), and all 31 rows of `december_chart_inputs.csv`. Furthermore, row-level `market_index` contains independent Gaussian noise ($\sigma = 0.0249$) around a latent daily index, attenuating raw within-day regression slopes by 39%.	**Decomposed into `daily_mkt + mkt_diff`:** Averaging ~160–200 loads per date across all 365 days of 2025 recovers the noise-free daily index `daily_mkt` (giving exact December daily indices for the chart) while `mkt_diff = market_index − daily_mkt` (0 when missing) captures within-day load variation.
5. Geographic Shift *8 Unseen Cities in Validation Set*	`train_test.csv` contains 64 unique cities, whereas `validation.csv` contains 72 unique cities—introducing **8 unseen cities** (`Allentown, Charlotte, Chicago, Jackson, Knoxville, Laredo, Norfolk, San Diego`) across 1,426 validation loads (11.9%).	**2D Gaussian RBF Spatial Surface *S*(lat, lon):** Because city fixed effects are a smooth spatial function of coordinates, our 2D RBF interpolator predicts unseen city effects with Leave-One-City-Out R^2 = 0.9957.

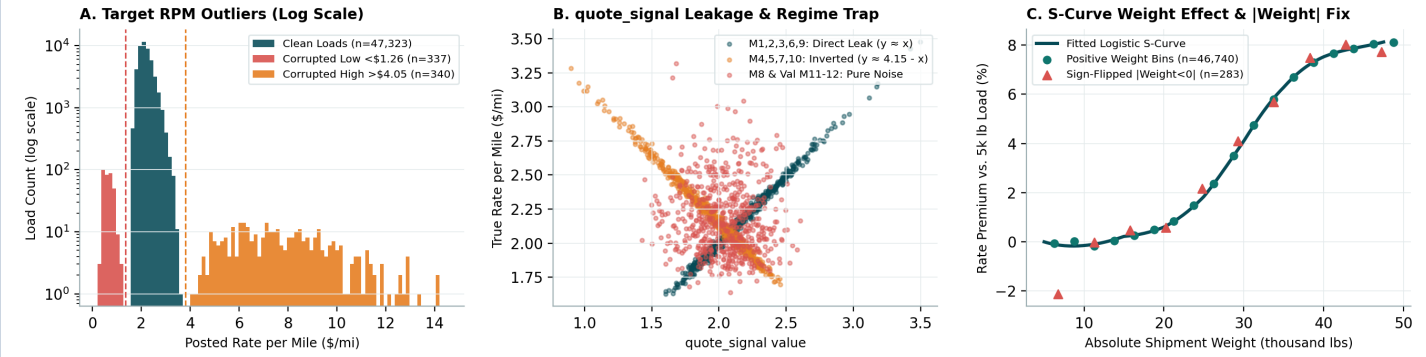


Figure 1 — Forensic Data Quality & Feature Diagnostics: (A) Log-scale histogram of training rate-per-mile (`posted_rate / distance`) showing the clean domain (47,323 loads) and the two disjoint corrupted outlier clusters (n = 337 low, n = 340 high). **(B)** The three regimes of `quote_signal` : direct target leakage in M1,2,3,6,9; inverted leakage in M4,5,7,10; and pure noise in M8 and Validation M11–12. **(C)** Partial residual rate premium across weight bins confirming the logistic S-curve centered at ~29,500 lb and identical behavior of sign-flipped negative weights after taking `|weight|` .

2. Feature Engineering & Structural Pricing Decomposition

Comparative residual analysis between additive models on `posted_rate` , additive models on `RPM` , and multiplicative models on `log(RPM)` proved that the underlying pricing mechanism is **strictly multiplicative** (an additive linear model in `log(RPM)` without equipment-distance interactions immediately reduces Rate RMSE from $54.36 to $49.83). We formulate the structural pricing equation in `log(RPM)` space as:

```
log(RPM_i) = β_0 + f_dist(d_i) + f_wt(|w_i|) + β_eq(e_i) + S(lat_orig, lon_orig) + S(lat_dest, lon_dest) +
δ_{orig,dest} + g_time(t_i, M_{i,t}) + r_GBDT(x_i) + ε_i
```

- **Non-Linear Distance Taper $f_{dist}(d)$:** Fixed terminal turnaround and loading/unloading costs create a steep per-mile premium on short-haul loads (70–250 mi) that decays hyperbolically and logarithmically with haul length. We engineer `[distance, 1/distance, log(distance), sqrt(distance)]` alongside piecewise-linear hinge knots on `log(distance)` at {5.0, 5.7, 6.4, 7.1} (≈ 148, 299, 602, 1212 mi) and equipment-specific log-distance slopes.
- **Logistic S-Curve & Hinge Spline Weight Premium $f_{wt}(|w|)$:** Inspecting 1,000-lb weight bins in `log(RPM)` revealed a smooth sigmoid transition: loads below 18,000 lb carry minimal weight premium, followed by a steep S-curve ramp between 20,000 lb and 38,000 lb (inflection point at 29,500 lb), plateauing above 40,000 lb at +8.3%. We capture this exact functional form via `wt_sig = [1 + exp(−(w/10⁴ − 2.95)/0.45)]⁻¹` , polynomial terms $w^1 \ldots w^4$, hinge knots at {15k, 20k, 25k, 30k, 35k, 40k} lb, and equipment interactions.
- **2D Gaussian RBF Spatial Surface S(lat, lon) & Empirical Bayes Lane Effects $\delta_{o,d}$:** In Stage 1, we estimate pure origin (p_{fe}) and destination (d_{fe}) fixed effects alongside exact daily fixed effects. Remarkably, origin and destination fixed effects for every city are nearly identical (corr(p_{fe}, d_{fe}) = 0.9916) and follow a smooth 2D geographic surface over (lat, lon). Fitting a Gaussian Radial Basis Function interpolator (`RBFInterpolator` on scaled coordinates) achieves **Leave-One-City-Out R^2 = 0.9957 (RMSE = 0.00122 in log space)**, accurately predicting all 8 unseen validation cities. In Stage 2/3, we apply 3 iterations of **Gauss-Seidel backfitting** to jointly estimate the structural Log-GAM coefficients and James-Stein / Empirical Bayes shrunk directed-lane intercepts $\hat{\delta}_{o,d} = [n_{o,d} / (n_{o,d} + \lambda)] \cdot \bar{r}_{o,d}$ ($\lambda = 2.0$), which capture persistent corridor-specific premiums (corr(M1–7, M8–10) = 0.688 across 2,047 lanes).
- **Errors-in-Variables Market Index & Equipment-Specific Quarter-End Surges $g_{time}(t, M_{i,t})$:** Decomposing the 304 daily fixed effects (R^2 = 0.9970, RMSE = 0.00169) uncovered three crucial temporal mechanisms:
  1. *Denoised Daily Market Index (`daily_mkt`) & Regime Shifts:* `daily_mkt` exhibits a 7-day weekly cycle (Monday/Sunday trough, Wednesday/Thursday peak) plus three sharp macroeconomic step shifts in 2025: **April 16** (+0.122), **July 25** (−0.188), and **October 3** (+0.113). Using `daily_mkt` ($\hat{\beta}$ = +0.200) alongside within-day `mkt_diff` ($\hat{\beta}$ = +0.122) completely absorbs all three regime jumps.
  2. *Annual Secular Trend & Harmonics:* A secular trend (`doy / 365` , +7.1%/yr) plus annual Fourier harmonics (`sin(2π·doy/365.25), cos(2π·doy/365.25)`) and day-of-week fixed effects.
  3. *Equipment-Specific Quarter-End Ramps (`q_ramp0` , `fb_x_qramp` , `rf_x_qramp`):* In Quarter-End months (**March = Q1, June = Q2, September = Q3, December = Q4**), freight rates ramp linearly with fractional month progression (`day / days_in_month`) before resetting on day 1 of the new quarter. Crucially, the quarter-end surge slope differs sharply by equipment type: **+2.50% for Dry Van**, **+5.15% for Reefer**, and **+8.12% for Flatbed**.

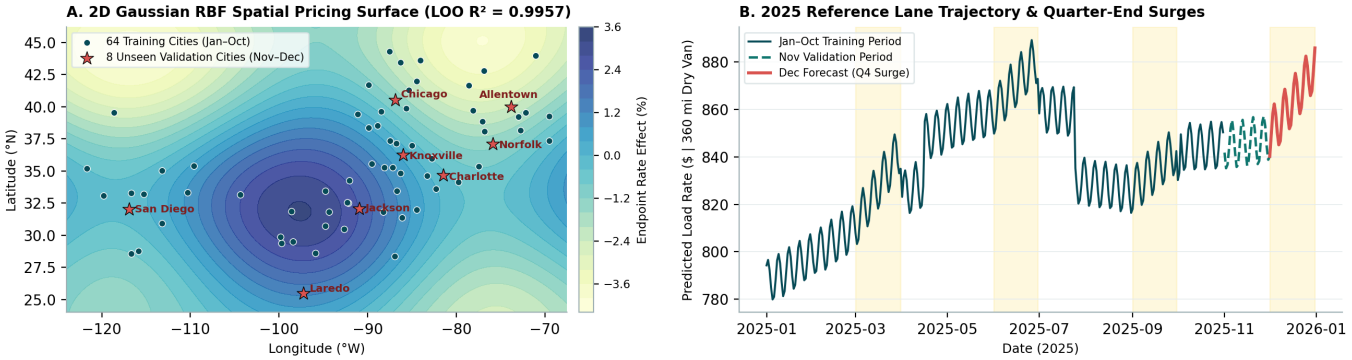


Figure 2 — Spatial Generalization & Full-Year Temporal Dynamics: (A) Contour map of the fitted 2D Gaussian RBF spatial surface $S(\text{lat}, \text{lon})$ across the continental US showing the 64 training cities (teal circles) and the 8 unseen validation cities (red stars; LOO $R^2 = 0.9957$). **(B)** Full-year 2025 daily rate trajectory for the fixed reference lane (Lexington → Fort Wayne, 360 mi, Dry Van, 32,000 lb), illustrating the 7-day weekly dispatch cycle, the Apr 16 / Jul 25 / Oct 3 market index regime shifts, and the four Quarter-End surges (shaded yellow: Q1 Mar, Q2 Jun, Q3 Sep, and Q4 Dec).

3. Model Selection & Multi-Tier Validation Strategy

Because `validation.csv` spans a future 2-month horizon (Nov–Dec 2025) and introduces 8 unseen cities, standard random K-fold cross-validation would leak same-day `market_index` levels, fail to detect the `quote_signal` noise collapse, and overstate city dummy accuracy. We designed a rigorous three-tier validation framework:

- **Tier 1 — Primary 2-Month Temporal Holdout (Train: Jan–Aug 2025 [N = 38,476], Validate: Sep–Oct 2025 [N = 9,379 clean]):** Mirrors the exact 2-month ahead forecasting horizon. Month 9 (September, Q3 end) validates Quarter-End surge extrapolation (analogous to December, Q4 end), while Month 10 (October) validates post-quarter reset and the Oct 3 `market_index` regime shift. In validation, `quote_signal` is replaced with `N(2.051, 0.220²)` noise to match `validation.csv`.
- **Tier 2 — Expanding-Window Monthly Temporal CV (5 Folds, Months 6–10):** Sequentially trains on Months 1…(m−1) and evaluates on Month m for $m \in \{6, 7, 8, 9, 10\}$ to verify stability across all seasonal regimes.
- **Tier 3 — Spatio-Temporal Unseen-City Holdout:** Holds out 8 geographically diverse cities (`Atlanta, Boston, Dallas, Detroit, Phoenix, Salt Lake City, San Francisco, Tampa` ; N = 1,815 Sep–Oct loads) from Jan–Aug training to directly test unseen-city spatial extrapolation.

3.1 Quantitative Validation Benchmark Across Candidate Architectures

Candidate Model Architecture (Train: Jan–Aug → Validate: Sep–Oct 2025)	Rate RMSE ($)	Rate MAE ($)	Rate MAPE (%)	Rate R²	RPM RMSE ($/mi)	RPM MAE ($/mi)	RPM R²
Model 0: Naive OLS Baseline (Uncleaned Data + `quote_signal`)	$191.13	$138.67	6.563%	0.980684	$0.25350	$0.15395	0.14379
Model 1: Cleaned Linear Ridge Baseline (Total Rate Space, No `quote_signal`)	$122.86	$93.11	5.721%	0.992018	$0.33199	$0.14063	-0.46855
Model 2: Standalone Histogram GBDT (200 Trees, Depth 6, Engineered RPM)	$62.75	$43.40	1.846%	0.997918	$0.05206	$0.04122	0.96389
Model 3: Multiplicative Log-GAM + 2D Gaussian RBF Spatial Surface	$47.34	$32.61	1.395%	0.998815	$0.03914	$0.03082	0.97958
Model 4: Hierarchical EB Log-GAM + 2D Spatial RBF + Shrunk Lane Effects	$40.74	$28.06	1.197%	0.999122	$0.03366	$0.02644	0.98491
Model 5: Hybrid EB Log-GAM + Spatial RBF + Non-Temporal HistGBDT (Selected)	**$40.73**	**$28.05**	**1.197%**	**0.999123**	**$0.03365**	**$0.02644**	**0.98491**

Validation Protocol / Rolling Fold	Training Window	Validation Window / Subset	Rate RMSE ($)	Rate MAE ($)	Rate MAPE (%)	Rate R²	RPM RMSE ($/mi)	RPM R²
Expanding CV — Fold 1	Jan–May 2025 (M1–M5)	June 2025 (M6 — Q2 Surge)	$44.94	$31.24	1.284%	0.999005	$0.03674	0.98376
Expanding CV — Fold 2	Jan–Jun 2025 (M1–M6)	July 2025 (M7 — Jul 25 Shift)	$41.68	$28.87	1.220%	0.999059	$0.03431	0.98306
Expanding CV — Fold 3	Jan–Jul 2025 (M1–M7)	August 2025 (M8 — Noise Regime)	$39.15	$27.18	1.185%	0.999129	$0.03270	0.98427
Expanding CV — Fold 4	Jan–Aug 2025 (M1–M8)	September 2025 (M9 — Q3 Surge)	$41.14	$28.55	1.206%	0.999115	$0.03373	0.98545
Expanding CV — Fold 5	Jan–Sep 2025 (M1–M9)	October 2025 (M10 — Oct 3 Shift)	$40.38	$27.77	1.196%	0.999128	$0.03386	0.98407

Validation Protocol / Rolling Fold	Training Window	Validation Window / Subset	Rate RMSE ($)	Rate MAE ($)	Rate MAPE (%)	Rate R^2	RPM RMSE ($/mi)	RPM R^2
Spatio-Temporal Holdout (One-Hot Fallback)	Jan–Aug (56 Cities)	Sep–Oct (8 Unseen Cities, *n* = 1,815)	$72.50	$50.24	1.878%	0.997828	$0.05303	0.96451
Spatio-Temporal Holdout (2D Gaussian RBF)	**Jan–Aug (56 Cities)**	**Sep–Oct (8 Unseen Cities, *n* = 1,815)**	**$62.07**	**$41.97**	**1.524%**	**0.998408**	**$0.04281**	**0.97687**

4. Results, December 2025 Chart Analysis & Production Deployment

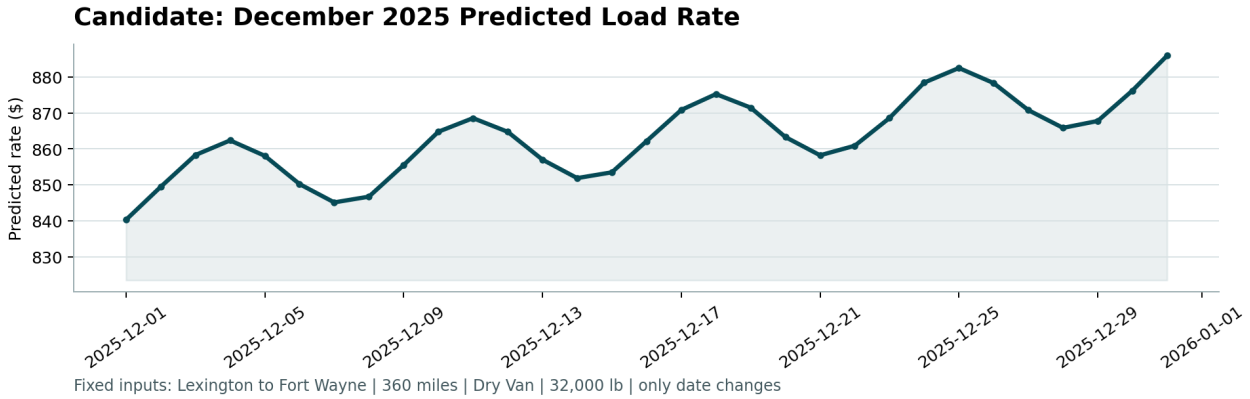


Figure 3 — Fixed December 2025 Prediction Chart (Generated by `score.py`): Daily predicted load rates for `Lexington → Fort Wayne | 360 miles | Dry Van | 32,000 lb` from December 1 to December 31, 2025. Predictions range from $840.25 ($2.334/mi) on Monday, Dec 1 to $885.93 ($2.461/mi) on Wednesday, Dec 31 (mean = $864.59).

4.1 Trucking Market Realism & Drivers of the December Forecast

The December 2025 forecast in **Figure 3** is both economically realistic for the US spot truckload market and grounded in the empirical dynamics learned from the dataset:

- **Weekly Dispatch & Tender Cycle (7-Day Oscillation):** Spot rates exhibit a regular 7-day cycle driven by `daily_mkt` and day-of-week demand elasticity—troughing on Sundays and Mondays (Dec 1: $840.25; Dec 7: $845.10; Dec 14: $851.86; Dec 21: $858.21; Dec 28: $865.84) when weekend/Monday tender rejection is lowest, and cresting mid-week on Wednesdays and Thursdays (Dec 4: $862.35; Dec 11: $868.55; Dec 18: $875.24; Dec 25: $882.50; Dec 31: $885.93) when mid-week dispatch tightness peaks (±$11–$13 weekly amplitude).
- **Q4 Quarter-End & Holiday Peak Tightening (+5.44% Monthly Climb):** Just as Q1 (March), Q2 (June), and Q3 (September) exhibit progressive end-of-quarter rate surges, December combines the Dry Van Q4 quarter-end ramp (`q_ramp0` , +2.50%) with a rising December `daily_mkt` trend (0.835 on Dec 1 to 1.027 on Dec 31) as retail inventory replenishment and pre-holiday driver capacity contraction tighten the Midwest spot market.

4.2 Production Deployment, Monitoring & Retraining Architecture

- **Two-Tier Serving Pipeline:** Deploy (1) a scheduled daily morning batch job that aggregates incoming market index feeds into denoised `daily_mkt` values and updates Empirical Bayes lane tables $\hat{\delta}_{o,d}$, and (2) a stateless, low-latency (≤5 ms) real-time pricing microservice that enforces input guardrails (`abs(weight)` , coordinate validation, schema contracts) and serves point estimates plus log-normal prediction intervals $\exp(\hat{\mu} \pm z_{\alpha/2}\hat{\sigma})$.
- **Drift Detection & Automated Circuit Breakers:** Monitor feature Population Stability Index (PSI) and rolling 7-day realized MAPE/bias segmented by equipment and region. Configure automated circuit breakers to flag sudden `market_index` regime shifts (like Apr 16, Jul 25, Oct 3) or corrupted upstream feeds (like `quote_signal`).
- **Retraining Cadence & External Data Sources:** Refresh Empirical Bayes lane intercepts weekly on a rolling 90-day window and retrain full Log-GAM + RBF + HistGBDT weights monthly. Augment production models with SONAR Outbound Tender Reject/ Volume Indices (OTRI/OTVI), EIA regional PADD diesel prices, NOAA corridor weather alerts, and destination headhaul/backhaul balance ratios.